

Vehicle Sales

Market Analysis Report

VehicleSales Table • Full Exploratory Analysis

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1. Dataset Overview

The data comes from the VehicleSales table inside the MotoDB SQL Server database. It covers used-car auction and retail transactions across the United States, with every row representing a single vehicle sale.

558,837 Raw Records before cleaning	17 Columns features per record	96 Car Brands unique makes	38 US States geographic coverage
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Column Reference

Column	Type	Description
Id	Integer	Unique record identifier
Year	Integer	Model year of the vehicle
Make	Text	Brand (Ford, Nissan, BMW ...)
Model	Text	Model name (F-150, Altima ...)
Trim	Text	Trim/class level (SE, LX, Limited ...)
Body	Text	Body style (sedan, SUV, coupe ...)
Transmission	Text	automatic / manual / unknown
VIN	Text	Vehicle Identification Number
State	Text	2-letter US state code
ConditionValue	Decimal	Vehicle condition score — normalised to 1–5
Odometer	Decimal	Mileage reading (log-transformed during cleaning)
Color	Text	Exterior paint color
Interior	Text	Interior color
Seller	Text	Selling entity / dealership name
MMR	Decimal	Manheim Market Report price (USD) — market benchmark
SellingPrice	Decimal	Actual transaction price (USD)
SaleDate	DateTime	Date and time of the sale

2. Missing Values — What Was Found & What Was Done

Before touching any data, we ran `df.isna().sum()` to get a full picture of the missing values. Here's what we found and the logic behind each decision:

2.1 VIN — 4 rows

Four rows had no VIN at all. On inspection, all four were Volkswagen Jetta entries with a body type of 'Navitgation' — clearly a data entry error for the entire record. These were dropped immediately since without a VIN there's no reliable way to identify the vehicle.

2.2 Make / Model / Trim / Body / Transmission — Up to 65,000 rows

This group was handled in two steps:

- Step 1 — WMI Decoding: The first 3 characters of the VIN (called the World Manufacturer Identifier) uniquely identify the car brand. A lookup table was built mapping WMI codes to their brands — for example WBA → BMW, SAL → Land Rover, 1FM → Ford. This alone recovered thousands of Make values.
- Step 2 — Group Mode Imputation: Whatever remained null was filled using the most common value within the same group. For example, if a row's Make was missing but its Model was known, it was filled with the most frequent Make for that Model across the whole dataset. Same logic applied to Trim and Body.
- Rows still null after both steps were dropped — these were vehicles with neither a decodable VIN nor any matching group to borrow from.
- Transmission nulls were filled with the label 'Unknown' rather than dropped, since 65,000 rows is too large to discard and the missing transmission doesn't affect most analyses.
- Body nulls remaining after imputation were also filled with 'Unknown'.

2.3 ConditionValue — 11,820 rows

Condition wasn't dropped or filled with a global average — instead the data was bucketed by Odometer into five ranges (New = under 20k miles, Good = 20k–60k, Average = 60k–120k, High = 120k–200k, Very High = above 200k). Each null ConditionValue was then filled with the median condition of vehicles in the same mileage bucket. The logic: a car with 150,000 miles should be assigned the condition typical for high-mileage cars, not the global median which mixes all categories.

2.4 Odometer, Color, Interior, MMR, SellingPrice, SaleDate

These columns had very few nulls (under 750 rows each). Odometer and price columns were filled with their median. Color and Interior were filled using group mode by Make/Model. Rows with missing SaleDate were dropped since the date is used for time-based analysis.

2.5 Final State After Cleaning

553,985 Clean Records after null handling	0 Remaining Nulls all columns complete	0 Duplicates fully unique rows
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3. Outliers — What Was Found & What Was Done

3.1 Odometer — The Most Complex Column

The raw Odometer ranged from 1 to 999,999 miles. Three separate issues existed:

Problem A — 999,999 as a Placeholder

Vehicles with exactly 999,999 miles were not cars with genuinely extreme mileage — their selling prices gave them away. A 2013 Hyundai selling for \$2,500 or a 2009 Honda Civic for \$1,000 are not high-mileage workhorses; 999,999 was simply what the system stored when the odometer reading was unknown. These rows were dropped.

Problem B — Extreme Right Tail (10,334 vehicles above 204,000 miles)

The IQR method placed the upper fence at ~204,000 miles. Removing 10,334 rows (~1.9% of data) would be wasteful. Instead, a log1p transformation was applied to the entire Odometer column. Log transformation compresses the long right tail without deleting any data or artificially capping real high-mileage values at the same number.

Problem C — Left Tail (vehicles under 1,000 miles)

After the log transformation, a cluster of near-zero values appeared on the left. These had original odometer readings of 1 mile — another placeholder used when the reading was unavailable. Their selling prices confirmed this (many under \$2,000 for relatively new models). These were dropped by filtering out any vehicle with fewer than 1,000 original miles. The two extreme right-side points above log(13.34) were also removed.

3.2 MMR & SellingPrice — Capped with IQR

Both price columns had ~16,000 outliers above their upper fence (~\$35,000). Rather than deleting these rows, the values were capped at the IQR upper bound. This means luxury vehicles like Bentley, Ferrari, and Tesla all show a capped MMR of ~\$34,950 — their true market values are higher, but for analysis purposes they sit at the ceiling rather than skewing distributions.

3.3 ConditionValue — No Treatment Needed

After normalization to the 1–5 scale, no outliers existed. The IQR bounds were -1.5 to 66.5, which means every legitimate value (1 through 5) fell comfortably within range.

3.4 Final Dataset After All Cleaning

551,409 Final Records ready for analysis	7,428 Rows Removed outliers & placeholders	17 Columns all retained
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4. Data Standardisation Before Analysis

Before moving to charts, several columns needed formatting fixes:

- **Body:** Values like 'Sedan', 'sedan', 'SEDAN' were all different entries for the same thing. Everything was lowercased and hyphens were replaced with spaces. Result: 87 messy unique values collapsed into 45 clean categories.
- **Transmission:** Lowercased. A bug was also caught — some Volkswagen entries had 'Sedan' stored as their Transmission value (a copy-paste error from the Body column). These were corrected to 'Unknown'.
- **Color:** The '-' symbol (used as a placeholder for unknown color) was replaced with the label 'Natural' to make it readable in charts.
- **ConditionValue:** Values were stored in two different scales — some 1–5, others 10–50 (a 10x version of the same scale). A normalisation function divided values above 5 by 10, bringing everything onto a consistent 1–5 scale.

New Columns Created for Analysis

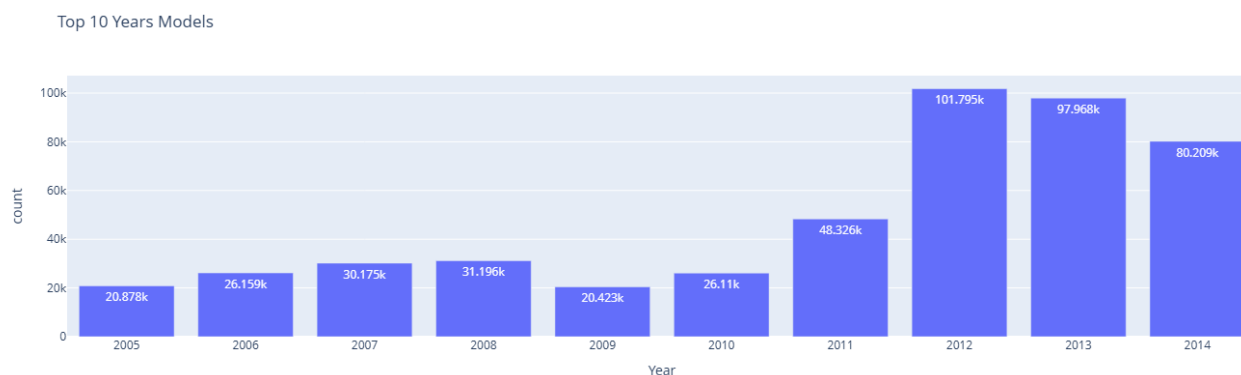
Column	How It Was Built	Used For
Odometer_Range	Bucketed Odometer into 5 bands	Condition imputation during cleaning
Car_Age	2026 minus the Year column	Depreciation & age analysis
Sale_Month	Extracted month from SaleDate	Seasonal pricing trends
Sale_DayOfWeek	Extracted weekday from SaleDate (0=Mon)	Day-of-week patterns
Price_Diff	SellingPrice minus MMR	Measures profit or discount vs. market
Price_Status	Categorised Price_Diff into 3 buckets	Profit/loss analysis across brands & months

5. Analysis & Findings

Everything from this point forward is about understanding the market. We'll go chart by chart — what was shown, what it tells us, and what's worth taking away.

5.1 Sales Volume by Model Year

A bar chart was plotted showing the top 10 model years by number of records in the dataset.

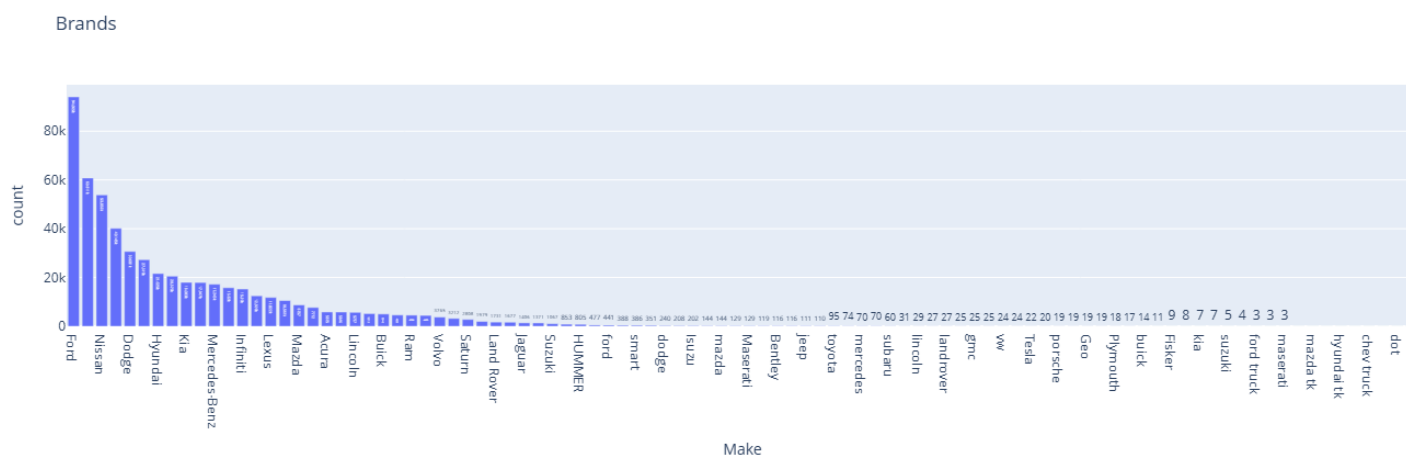


📌 **What this means:** Models from 2011–2014 make up over 60% of the dataset. This isn't coincidence — it reflects the lifecycle of used cars. Cars sold at auction or through dealers typically arrive when they're 3–5 years old, which means a dataset captured around 2015–2017 would be dominated by 2011–2014 models. The drop after 2014 makes sense too: fewer people trade in cars that are only 1–2 years old.

📌 **The 2008–2010 dip:** 2008 and 2009 are present but lower than the years immediately before and after them. This likely reflects the aftermath of the 2008 financial crisis — fewer new cars were sold in those years, which means fewer used cars from those years were available later.

5.2 Brand Distribution

A bar chart showing all 96 brands sorted by volume, and individual charts per brand for models.

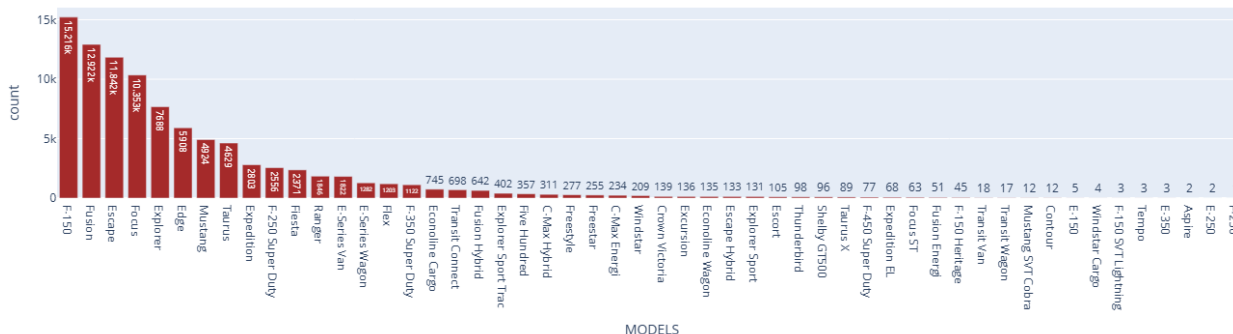


📌 **Ford's dominance:** Ford alone accounts for 17% of the entire dataset — more than Chevrolet and Nissan combined. This reflects Ford's historically strong market share in the US, especially driven by the F-150 pickup truck which is the best-selling vehicle in America for decades running.

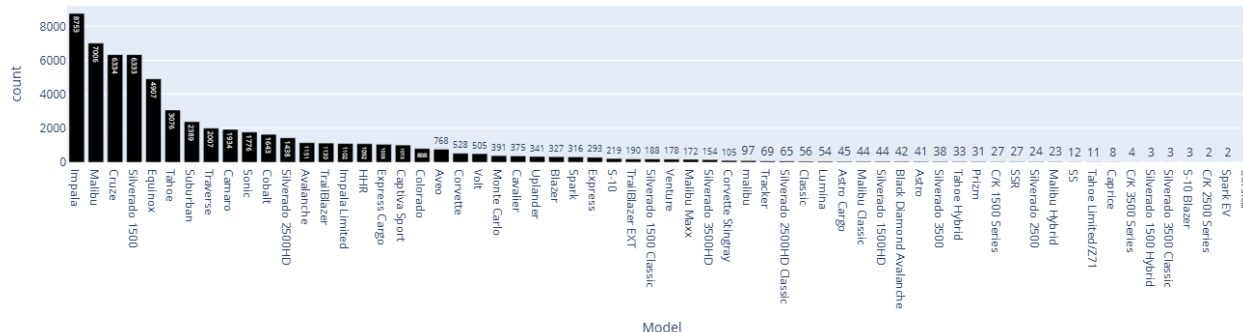
📌 **American brands lead, but Japanese brands are strong:** The top 4 spots are split between American (Ford, Chevrolet, Dodge) and Japanese (Nissan, Toyota, Honda) brands. European brands like BMW and Volkswagen appear but at significantly lower volumes, consistent with their premium/niche positioning in the US market.

Top Models per Brand

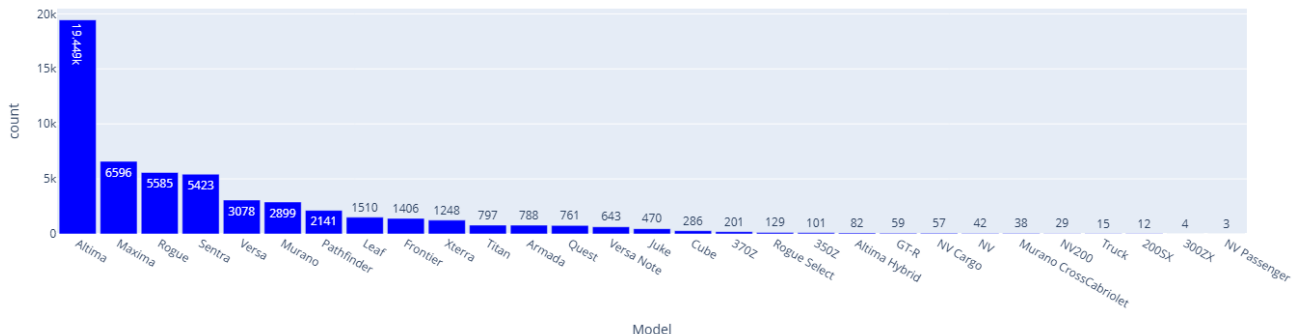
Ford Models

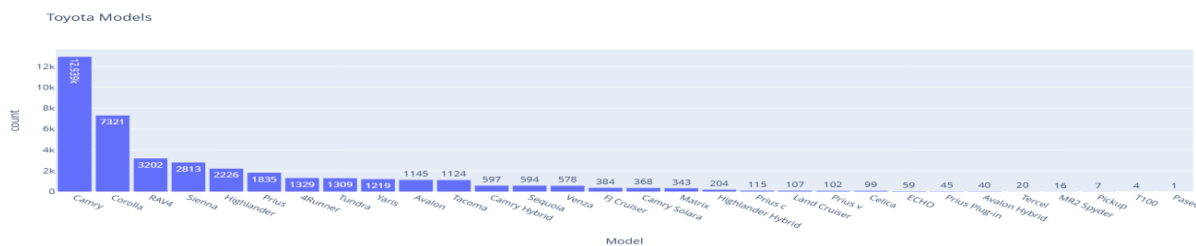


Chevrolet Models



Nissan Models

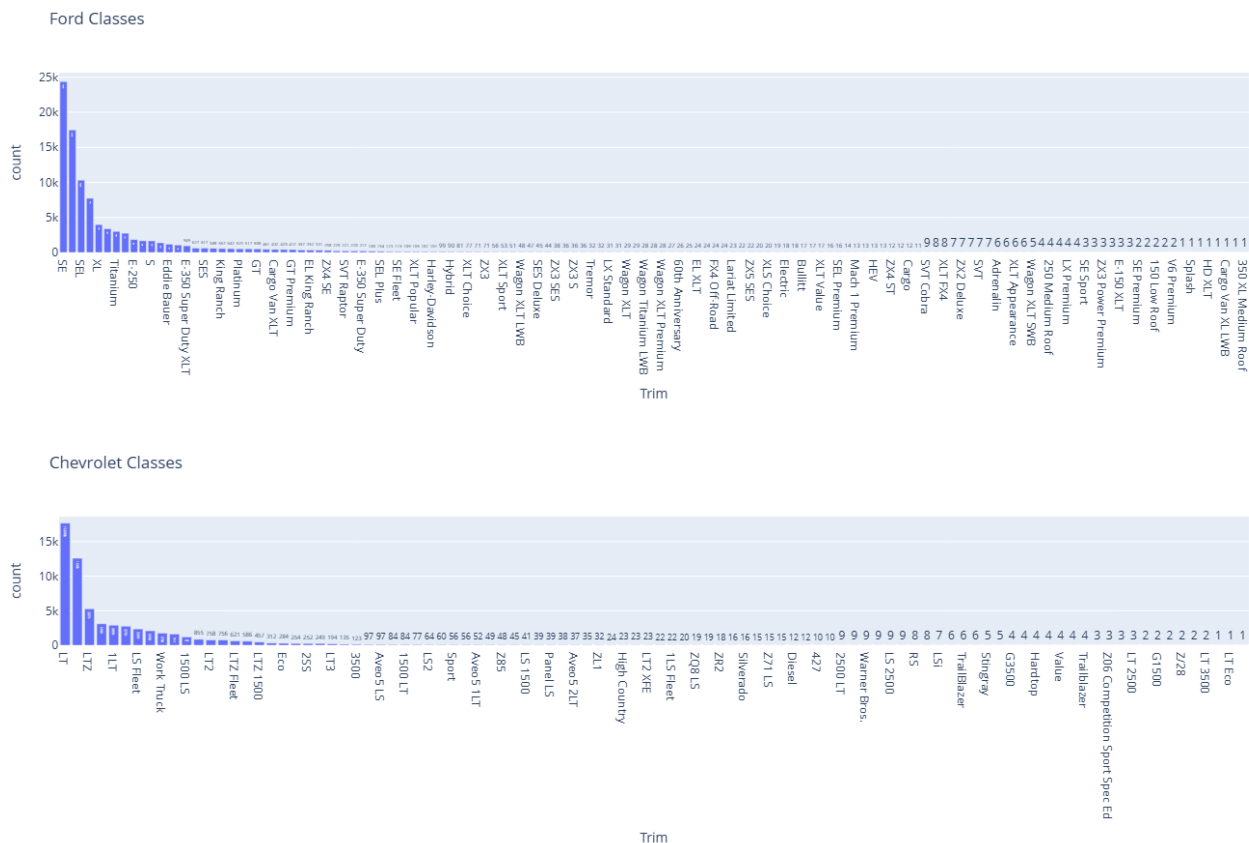




📌 **Trucks & SUVs vs. sedans:** Ford's top model is the F-150 (truck), Chevrolet's is the Silverado (truck), while Nissan and Toyota are led by sedans. This reflects the broader US market split — American brands dominate trucks, Japanese brands dominate sedans.

5.3 Trim Levels by Brand

Bar charts were produced for Ford, Chevrolet, Nissan, Dodge, Honda, and BMW showing the distribution of trim levels.



📌 **Entry and mid-level trims dominate:** Across all brands, the base and mid-level trims (SE, LX, LT, S, SXT) account for the majority of volume. Top-tier trims (Limited, Platinum, Sport) appear but are much less frequent. This makes sense in a used-car context — buyers looking for value gravitate toward the more affordable trim levels.

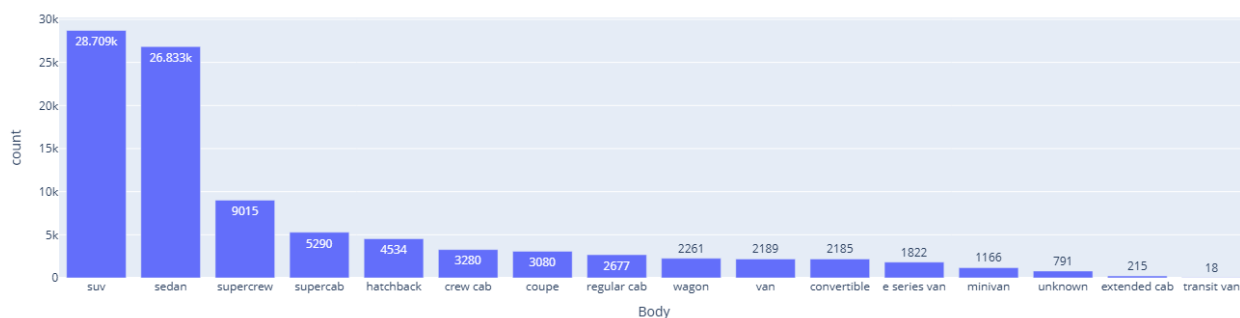
✦ **Ford SE vs XLT gap:** Ford's SE has 24,372 records vs XLT's 17,474. SE is primarily used on passenger cars (Fusion, Focus, Escape) while XLT is an F-Series trim. The fact that SE leads tells us Ford's passenger cars collectively outsell the F-Series in terms of variety, even though the F-150 alone is the single most-recorded model.

5.4 Body Style Distribution

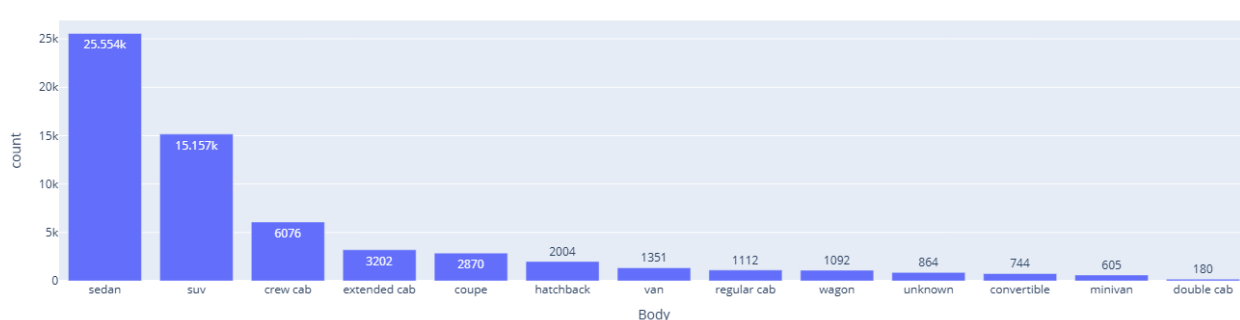
After standardising Body to lowercase and replacing hyphens with spaces, 45 unique body styles remained. Bar charts were created for Ford, Chevrolet, Nissan, Dodge, and BMW.

Body Style	Count	Share
sedan	240,169	43.6%
suv	143,287	26.0%
hatchback	26,048	4.7%
minivan	25,399	4.6%
coupe	17,609	3.2%
crew cab	16,347	3.0%
wagon	16,064	2.9%
convertible	10,409	1.9%
supercrew	9,021	1.6%
Other (36)	~47,056	8.5%

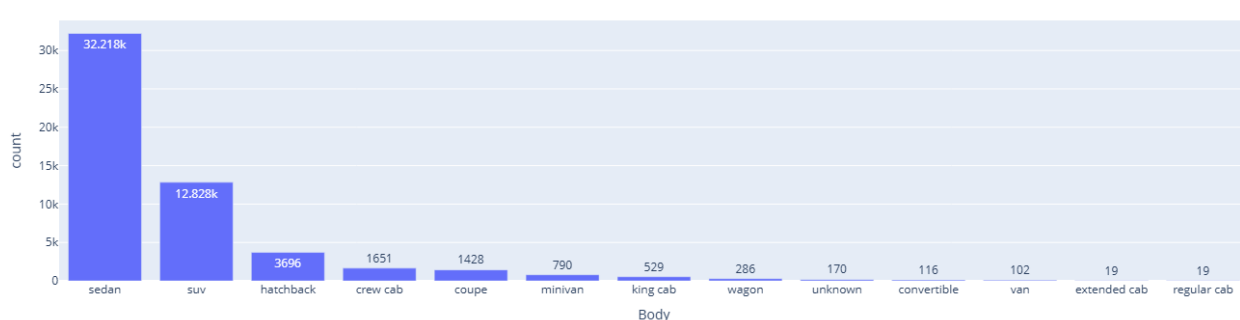
Ford Shape Types



Chevrolet Shape Types



Nissan Shape Types



📌 **Sedans are king — but SUVs are close:** Sedans account for 43.6% of the market while SUVs hold 26%. Together they cover nearly 70% of all vehicles. However, the gap between sedans and SUVs in this dataset (which mostly covers 2005–2014 models) would be significantly smaller in a more recent dataset, as SUVs have been rapidly gaining market share since the mid-2010s.

📌 **Brand personality through body style:** Ford and Chevrolet show heavy truck/SUV presence (crew cab, supercrew, SUV). Nissan and Toyota lean toward sedans. BMW is almost entirely sedans and coupes, reflecting their passenger-car focus. Dodge is interesting — the Grand Caravan makes minivan a significant category for them specifically.

5.5 Transmission

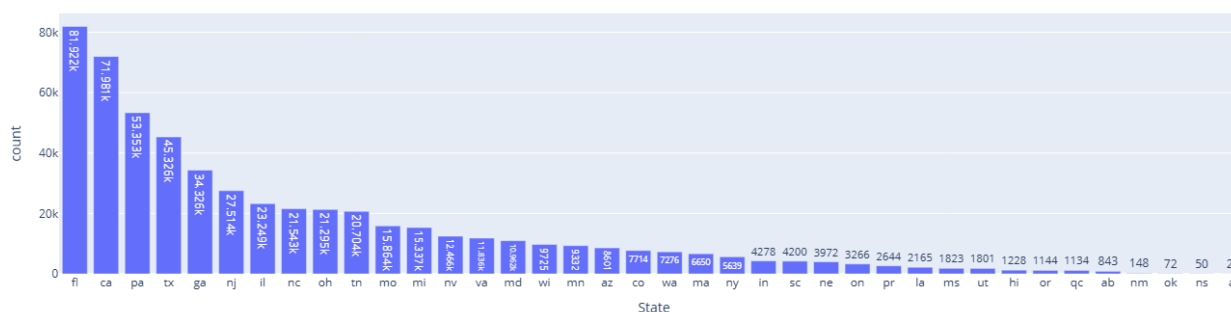
Transmission	Count	Share
Automatic	534,053	96.9%
Manual	17,305	3.1%
Unknown	51	0.01%

📌 **Manual transmissions are nearly extinct in this market:** Only 3.1% of vehicles have manual transmission. This aligns with US market trends where automatics have dominated since the 1990s. The manual cars present are likely sports cars (Mustang, BMW M-series, Mazda MX-5) or older entry-level economy cars where manual was the standard option.

5.6 Geographic Distribution — Top States

Rank	State	Records	Why it's high
1	FL (Florida)	81,922	Large population, warm climate, active car market
2	CA (California)	71,981	Largest US state by population
3	PA (Pennsylvania)	53,353	Major auction facilities in the region
4	TX (Texas)	45,326	Second largest state, strong truck market
5	GA (Georgia)	34,326	Manheim Atlanta — one of the largest auto auctions in the world
6	NJ (New Jersey)	27,514	Dense population, proximity to major metros
7	IL (Illinois)	23,249	Chicago metro market
8	NC (N. Carolina)	21,543	
9	OH (Ohio)	21,295	
10	TN (Tennessee)	20,704	

Top 10 States



📌 **Florida leads — likely because of auction concentration:** Florida and California together represent nearly 28% of all records. Florida in particular is home to major Manheim auction facilities and has a uniquely active

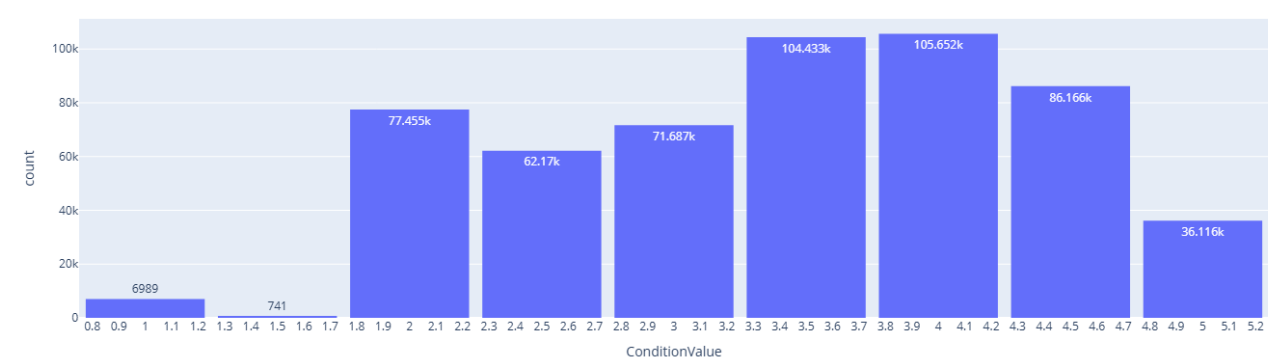
used-car market driven by a large retiree population, high vehicle turnover, and year-round driving conditions with minimal rust damage.

📌 **Georgia stands out relative to its size:** Georgia (ranked 8th by US population) is 5th in this dataset. This points directly to the Manheim Atlanta auction, which processes hundreds of thousands of vehicles annually and is one of the highest-volume auto auctions in North America.

5.7 Vehicle Condition

ConditionValue was stored inconsistently — some values on a 1–5 scale, others on a 10–50 scale (same scale, multiplied by 10). A normalisation function fixed this, then a histogram was plotted.

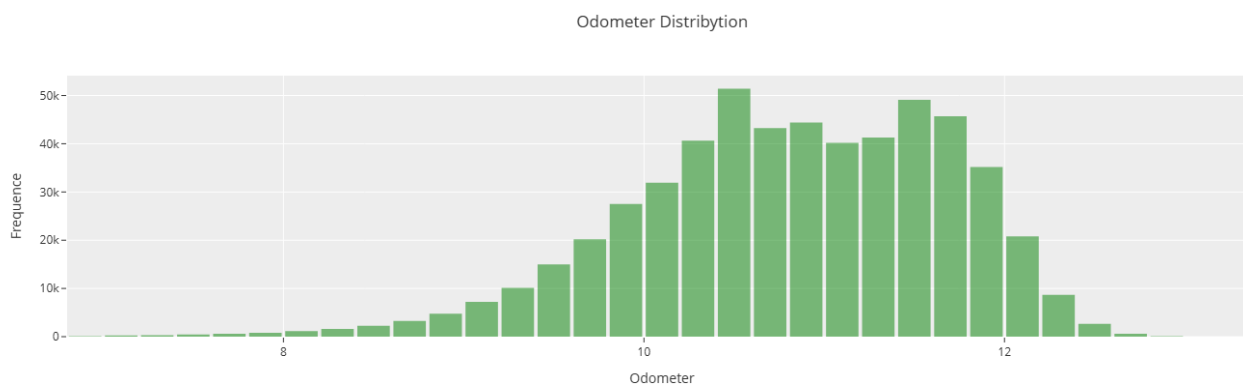
Statistic	Value
Mean	3.42 / 5.0
Median	3.60 / 5.0
Std Dev	0.94
25th %	2.7
75th %	4.2
Min	1.0
Max	5.0



📌 **Most cars are in above-average condition:** The median condition is 3.6 out of 5 — above the midpoint. The distribution is slightly left-skewed, meaning more vehicles cluster toward the 3–5 range than the 1–2 range. This makes sense for an auction/dealer dataset — severely damaged vehicles are typically sent to salvage auctions, not mainstream used-car markets.

📌 **Condition 1.0 vehicles still appear:** The minimum of 1.0 indicates the dataset does include some rough vehicles. These are likely high-mileage or accident-affected cars that still have enough value to sell through mainstream channels.

5.8 Odometer Distribution (After Log Transformation)

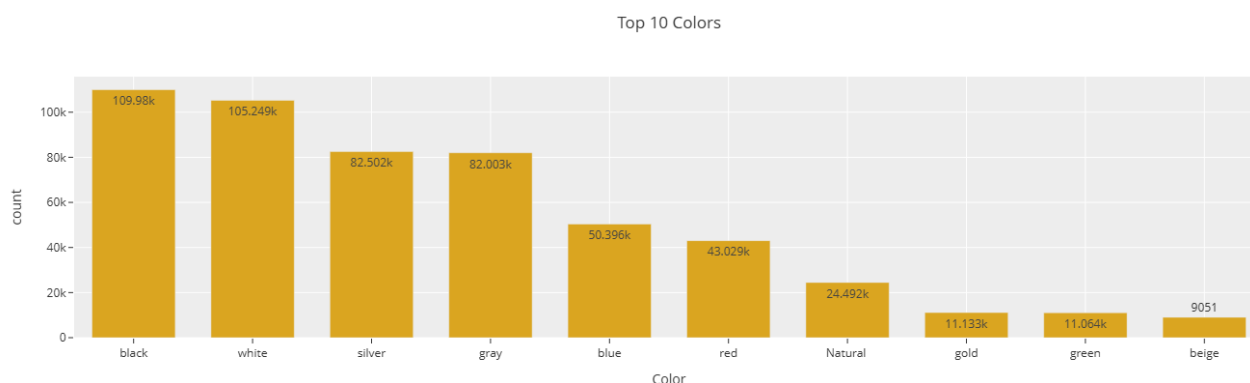


A histogram with 47 bins was plotted on the log-transformed Odometer column. The x-axis now shows log values (roughly 8.4 to 13.3), which translate to approximately 4,400 to 600,000 original miles.

📌 **The distribution is now approximately normal:** After the log transformation, the heavily right-skewed raw distribution became roughly bell-shaped. The bulk of the data sits between $\log(10.25)$ and $\log(11.5)$, which corresponds to about 28,000 to 98,000 original miles — exactly the range you'd expect for used cars 3–10 years old.

📌 **Why log transformation matters:** If we had left Odometer as-is, any regression model or correlation analysis would be distorted by the extreme values. The log scale treats the difference between 1,000 and 10,000 miles as equivalent in importance to the difference between 10,000 and 100,000 miles — which is economically the correct way to think about mileage impact on value.

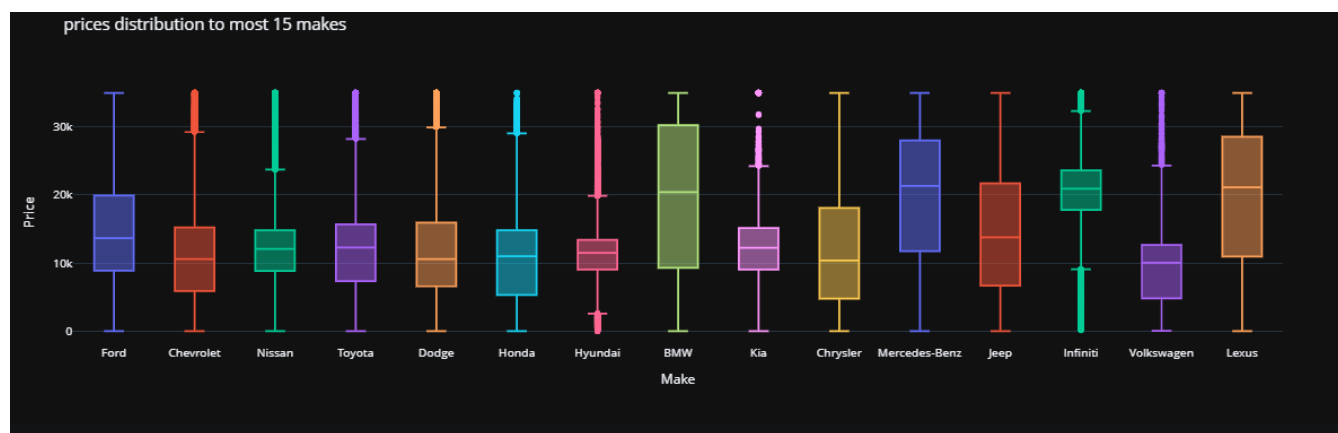
5.9 Exterior Color



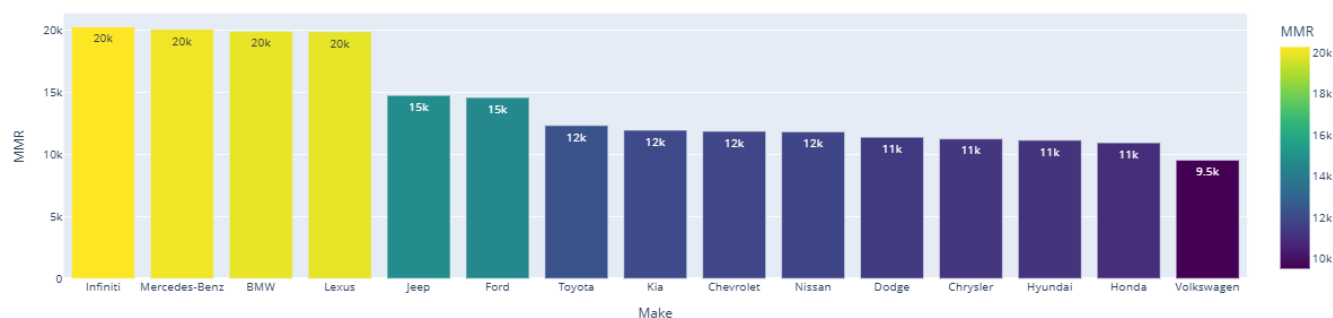
📌 **Neutral colors dominate the market completely:** Black, White, Silver, and Gray together account for nearly 69% of all vehicles. This isn't a coincidence — these colors have consistently higher resale value because they appeal to the broadest range of buyers. Dealers and fleet operators (who represent a large share of auction inventory) specifically order these colors to minimize resale risk.

📌 **Bold colors are rare but present:** Red sits at 7.8% — the most popular 'statement' color. Orange, Purple, and Yellow together account for under 1%. These colors are typically chosen by individual buyers rather than fleets, and they tend to have longer time-to-sell and lower resale values.

5.10 MMR Price Distribution by Brand



Average MMR prices for top brands



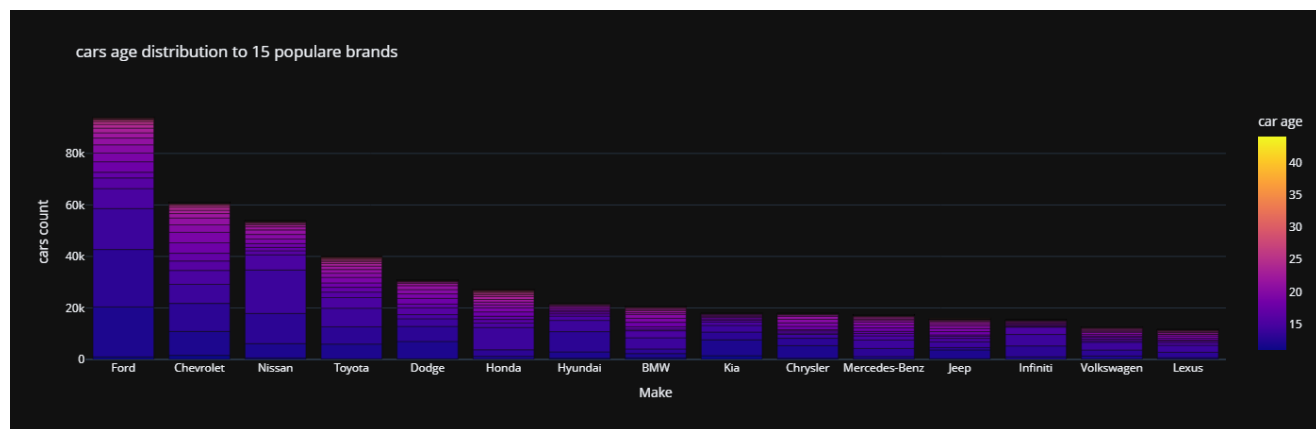
A box plot was created for the top 15 brands by volume, showing the spread of MMR (Manheim Market Report) values.

📌 **MMR is the market's pricing anchor:** MMR represents what the industry believes a vehicle is worth at wholesale. The box plot reveals enormous variation between brands. Luxury brands (BMW, Mercedes-Benz, Land Rover) show higher median MMR and wider boxes (more price variability). Volume brands (Ford, Nissan, Chevrolet) show tighter distributions centered around \$10,000–\$18,000.

📌 **The \$34,950 ceiling effect:** Many luxury brands (Aston Martin, Bentley, Ferrari) show a flat median at exactly \$34,950 — this is the IQR cap from outlier treatment. Their actual market values are much higher, but for analysis consistency they were capped. Any modelling work should treat these brands separately.

📌 **Ford and Chevrolet have wide ranges despite being volume brands:** This reflects the diversity of their lineups — a Ford Fiesta and a Ford F-350 Super Duty are both in the dataset and command very different prices. The wide box isn't noise; it's product diversity.

5.11 Car Age Distribution by Brand



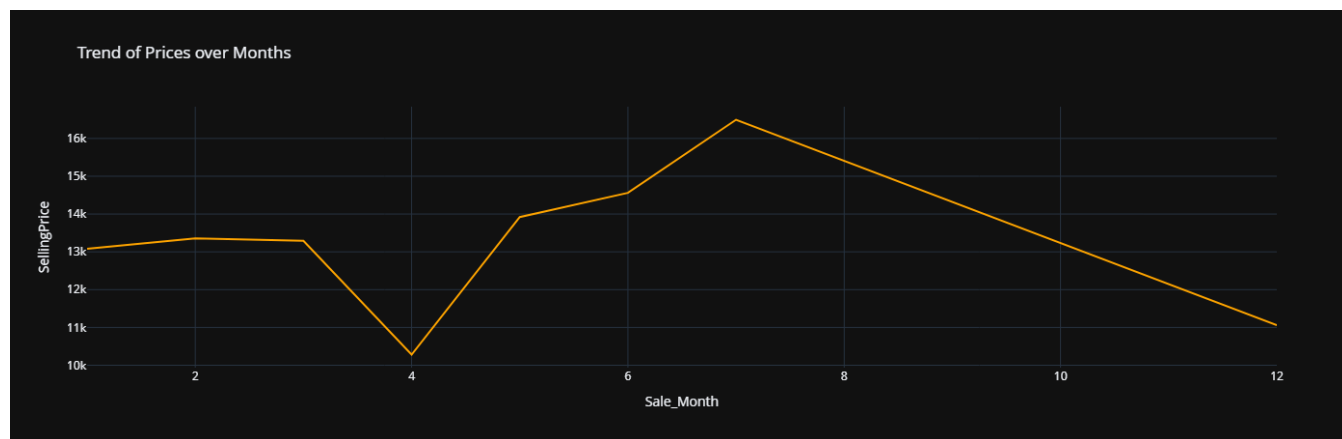
A stacked bar chart showed the age mix for the top 15 brands — how many cars of each age (2026 minus model year) each brand contributed.

✦ **Most vehicles are 12–15 years old:** Since the dataset covers mostly 2011–2014 model years and we're in 2026, the typical car is 12–15 years old. Ford and Chevrolet show the broadest age range, reflecting their large volume across all years.

✦ **Newer models (age 3–7) are relatively rare:** Fewer 2019–2023 models appear. This could reflect the data collection cutoff or a lag between when cars are sold new and when they enter the used/auction market at scale.

5.12 Average Selling Price by Month

A line chart traced the average SellingPrice for each calendar month.



✦ **Relative Stability (January - March):** The first three months of the year (January, February, and March) saw very stable prices, fluctuating within a narrow range around 13,000. This indicates that supply and demand were in equilibrium at the beginning of the year.

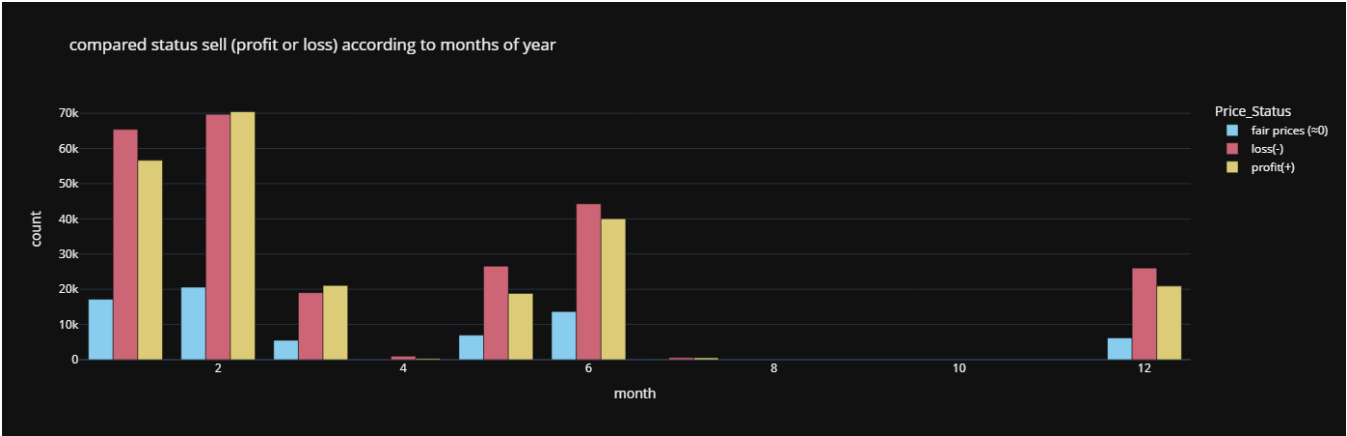
✦ **The Sharp Drop in April:** This is the most important point in the chart; there was a sudden and significant price drop in April, reaching its lowest point (approximately 10,200).

Possible explanations: There might have been a temporary slowdown, a large influx of inexpensive cars into the market, or perhaps significant sales and discounts during this period.

- ✦ **The Rocket Rise (May - July):** After April, prices began a sharp and continuous rise for three consecutive months. The price peak was reached in July, with an average price approaching 16,500. This often represents the "peak season," and may be linked to the summer or holiday season, when demand for cars increases.
- ✦ **Gradual Decline (August - December):** From July until the end of the year, there is a clear and continuous downward trend. Prices dropped in December to around \$11,000, the second lowest price of the year after April.

5.13 Price Status — Profit, Loss, or Fair

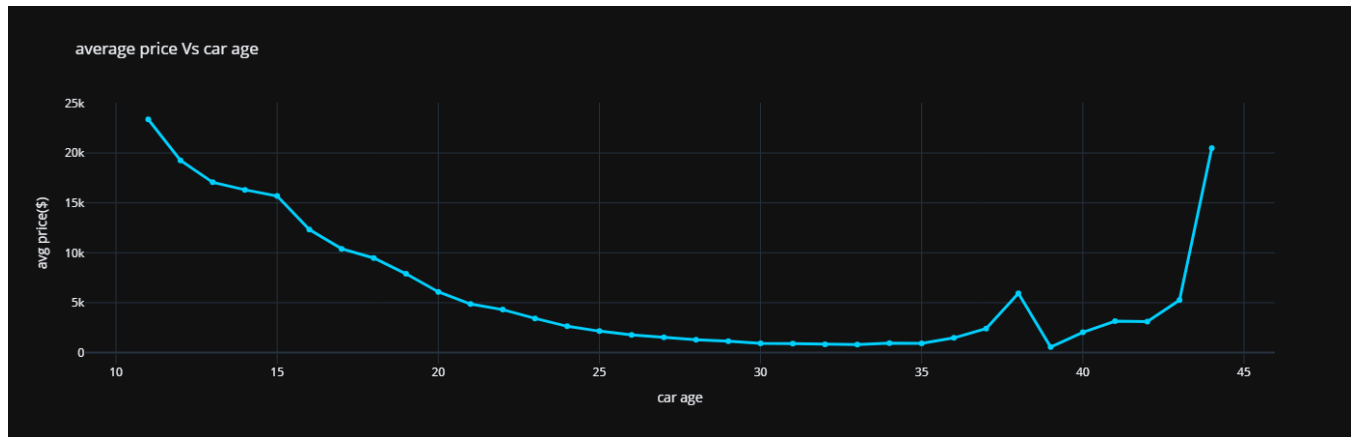
Price_Diff = SellingPrice – MMR was calculated. If the difference exceeded +\$100 the sale was 'profit(+)', below -\$100 was 'loss(-)', and within ±\$100 was 'fair prices (≈0)'. A pie chart and grouped bar chart by month were produced.



- ✦ **Sellers mostly sell below market value (loss):** Across all months except February, loss transactions outnumber profit transactions. This is typical for wholesale auctions — sellers (rental companies, banks, dealers) prioritise speed of sale over maximising price. They accept below-MMR offers to clear inventory quickly.
- ✦ **February is the exception:** February is the only month where profit transactions narrowly edge out loss transactions (70,435 vs 69,700). This aligns with the price trend finding — February demand is strong enough that sellers can hold firm on price.
- ✦ **Fair-priced transactions are the minority:** Transactions that land within ±\$100 of MMR represent a surprisingly small share. Most transactions deviate meaningfully from the market benchmark, suggesting that MMR is a reference point rather than a fixed price — actual negotiations move the final price significantly in either direction.

5.14 Average Price vs. Car Age

A line chart with markers plotted the average SellingPrice for each Car_Age value (from 1 to 20+ years).

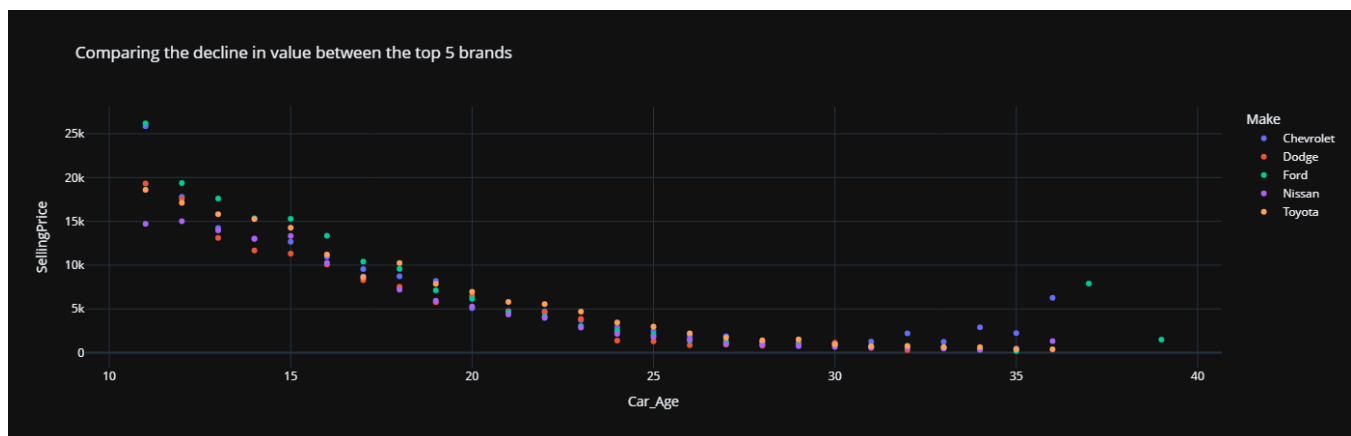


✦ **Clear depreciation curve:** The line shows a steep drop in the first 3–5 years of a car's life, then a more gradual decline. A car loses the most value in its earliest years — this is the well-known 'new car depreciation cliff' playing out in the data.

✦ **The curve flattens for older vehicles:** Cars older than 10–12 years show relatively stable (and low) average prices. At some point the vehicle has depreciated as much as it's going to and further age has diminishing impact on price.

5.15 Depreciation Comparison — Top 5 Brands

A scatter plot overlaid the depreciation curves for Ford, Chevrolet, Nissan, Toyota, and Dodge.



✦ **Toyota holds value best:** Toyota's depreciation curve sits consistently above the others at all age points. A 7-year-old Toyota commands a higher average price than a 7-year-old Ford or Chevrolet of the same age. This reflects Toyota's reputation for reliability and lower long-term ownership costs.

✦ **Ford and Chevrolet converge with age:** Both American brands start strong but their values drop faster. By 10+ years old, the gap between brands narrows significantly — at that price point buyers care less about brand and more about condition and mileage.

✦ **Dodge depreciates fastest:** Dodge shows the steepest early-year decline. This is consistent with industry data — Dodge vehicles tend to have higher initial MSRP incentives and lower long-term demand in the used market.

5.16 Condition vs. Price Status

A box plot compared ConditionValue across the three Price_Status groups (profit, loss, fair).



✦ **Better condition slightly increases profit likelihood:** Vehicles sold at a profit have marginally higher median condition scores than those sold at a loss. However the overlap between groups is large — condition alone is not enough to determine whether a car will sell above or below MMR.

✦ **The spread is widest for fair-priced vehicles:** Cars sold at approximately MMR value show the widest condition range — from 1.0 to 5.0. This suggests that at the fair price point, a wide range of condition levels can find a matching buyer, whereas profit sales tend to cluster in better-condition vehicles and loss sales in worse-condition ones.

5.17 Average Selling Price by US State (Choropleth Map)

A choropleth map colored US states by their average SellingPrice.

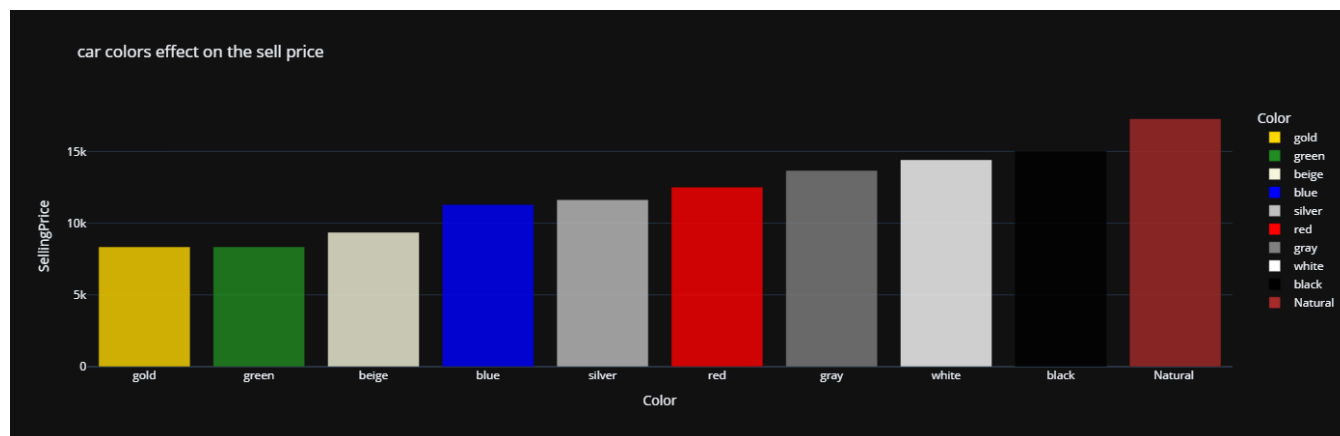


✦ **Western and Northern states show higher average prices:** States like California, Washington, and the Northeast tend to show darker shading (higher prices). This likely reflects a combination of higher income levels, stricter emissions standards that favor newer/better-maintained vehicles, and higher general cost of living.

✦ **Southern states show lower averages:** States in the Southeast show lighter shading. This is consistent with a higher proportion of older, higher-mileage vehicles in those markets and generally lower vehicle price points.

5.18 Average Selling Price by Color

A bar chart showed the average SellingPrice for the top 10 colors, with each bar colored to match the actual car color.

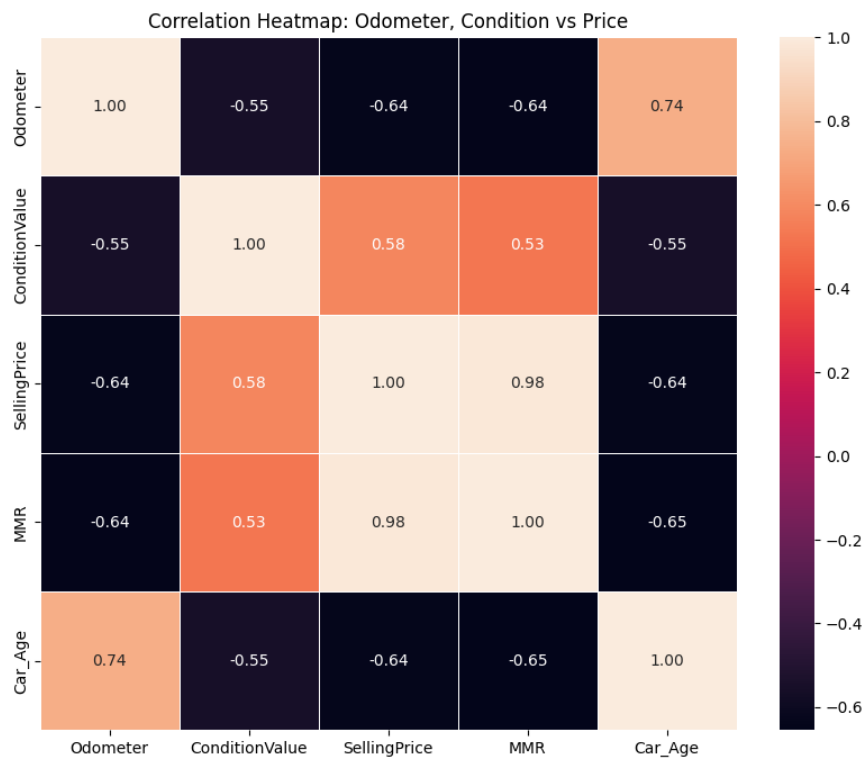


✦ **White and silver command slight premiums:** White and silver vehicles show marginally higher average selling prices than black or gray. This could reflect a composition effect — white and silver are common in fleet vehicles (which tend to be newer and lower mileage) more than black, which is popular in both fleets and consumer vehicles.

✦ **'Natural' (unknown color) prices are lower:** The 'Natural' category — which represents vehicles where color wasn't recorded — shows below-average prices. Missing data often correlates with older or damaged vehicles where the paperwork is incomplete.

5.19 Correlation Analysis — Two Heatmaps

First Heatmap: Odometer, Condition, SellingPrice, MMR, Car Age

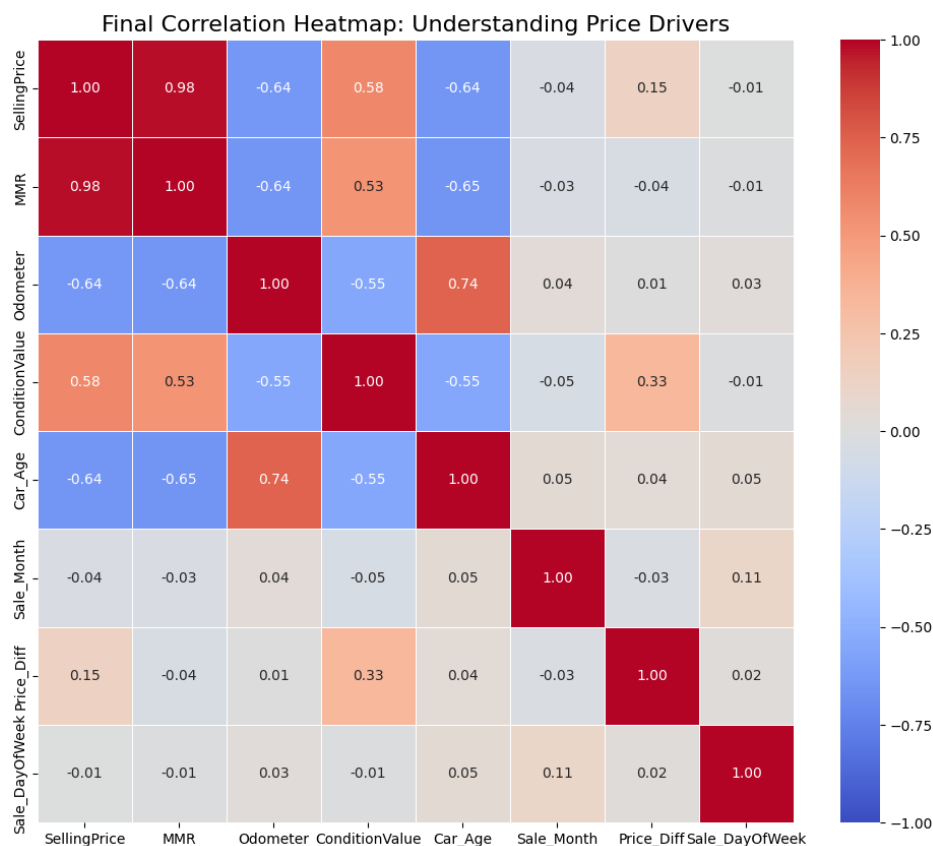


Feature Pair	Direction	Strength	What it means
SellingPrice ↔ MMR	Positive	Very Strong	MMR is the best predictor of actual price
SellingPrice ↔ Car_Age	Negative	Moderate	Older cars sell for less
SellingPrice ↔ Odometer	Negative	Moderate	More miles = lower price
SellingPrice ↔ ConditionValue	Positive	Moderate	Better condition = higher price
Odometer ↔ Car_Age	Positive	Strong	Older cars have accumulated more miles
Odometer ↔ ConditionValue	Negative	Moderate	Higher mileage = worse condition

📌 **MMR is the anchor of the entire pricing system:** The very high positive correlation between SellingPrice and MMR confirms that Manheim's market reports are genuinely reliable. Sellers and buyers use MMR as their reference point, and actual transactions cluster around it. This is the single most important relationship in the dataset.

📌 **Condition, age, and mileage all matter but don't dominate:** All three have moderate correlations with price — meaningful but not overwhelming. This tells us that no single factor drives price; it's a combination. A high-mileage car in excellent condition can still command a good price; a low-mileage car in poor condition will be discounted.

Second Heatmap: Full Feature Set (Including Sale_Month, Sale_DayOfWeek, Price_Diff)

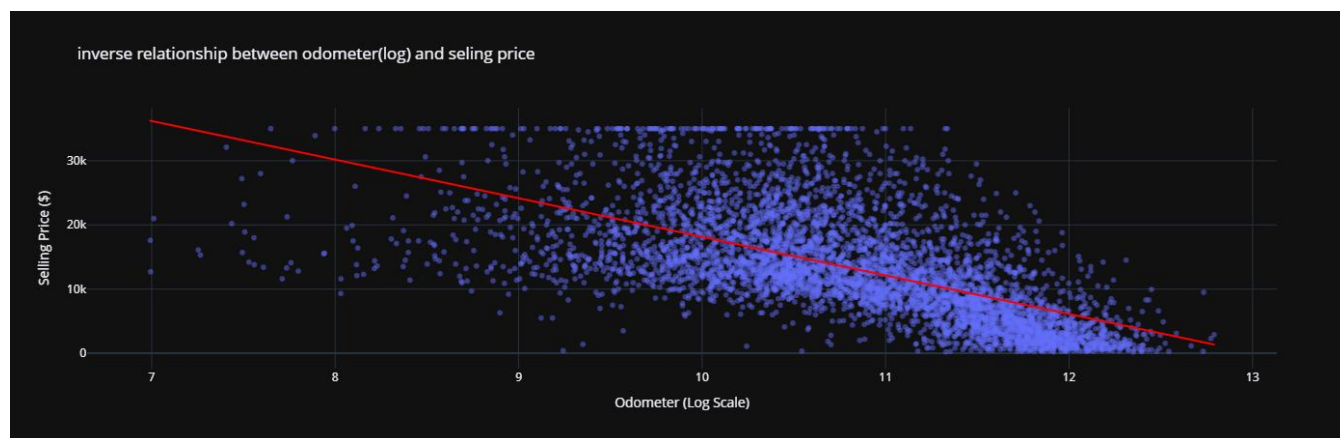


✦ **Timing has almost no correlation with price:** Sale_Month and Sale_DayOfWeek show near-zero correlations with SellingPrice and MMR. Whether a car is sold in January or March, on a Monday or a Thursday, has negligible impact on the final price. This debunks the common belief that timing a sale carefully produces better prices.

✦ **Price_Diff is mechanically correlated with SellingPrice:** Since $\text{Price_Diff} = \text{SellingPrice} - \text{MMR}$, it's mathematically expected that Price_Diff will correlate with SellingPrice. The interesting finding is that Price_Diff has a negative correlation with MMR — lower-priced (cheaper) vehicles tend to sell at a larger discount below MMR, while expensive vehicles are more likely to sell at or above their market benchmark.

5.20 Odometer vs. Selling Price Scatter (with OLS Trendline)

A sample of 5,000 points was plotted with SellingPrice on the y-axis and $\log(\text{Odometer})$ on the x-axis. A red OLS (ordinary least squares) regression line was overlaid.

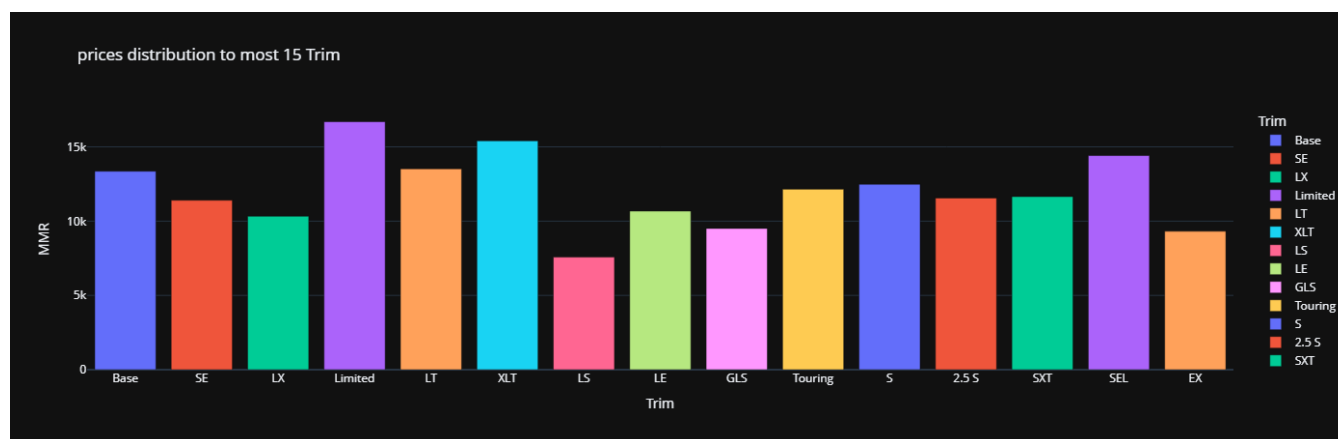


✚ **Clear negative relationship confirmed visually:** The red trendline slopes downward — as $\log(\text{Odometer})$ increases, SellingPrice decreases. This is what economic theory predicts and the chart confirms it cleanly.

✚ **Wide scatter means odometer is necessary but not sufficient:** Despite the clear trend, individual points are spread widely around the trendline. A car with $\log(\text{Odometer}) = 11$ could sell anywhere from under \$5,000 to over \$25,000. This means other factors — brand, model, condition, trim, color — add substantial independent explanatory power beyond mileage alone. A good pricing model needs all of these inputs together.

5.21 MMR by Trim Level — Top 15 Trims

The top 15 most common trim levels were compared by their average MMR in a bar chart.



✚ **Trim level directly maps to price tier:** Limited, Titanium, and Lariat command the highest average MMR among common trims, while Base, S, and L sit at the bottom. This confirms that trim level is a reliable proxy for vehicle spec and price — something any pricing model should incorporate.

✚ **The SE trim is an outlier — it covers multiple brands:** SE is used by Ford, Nissan, Hyundai, and many others, so its average MMR reflects a blend of different vehicles and price points rather than a single consistent spec level. Brand-specific trim analysis is more meaningful.

6. Key Takeaways

Ford Top Brand 94,065 records — 17.1%	2012 Top Model Year 101,795 records	Sedan Top Body Style 43.6% of all vehicles	Florida Top State 81,922 records
3.42 Avg Condition out of 5.0 — above average	96.9% Automatic manual nearly gone	Black Top Color 19.9% of all cars	Feb Best Month highest prices & profit rate

Summary

- The dataset is dominated by 2011–2014 model years, reflecting the natural lifecycle of used cars entering the auction market 3–5 years after new-car purchase.
- Ford, Chevrolet, and Nissan together account for 38% of the dataset. American brands own trucks; Japanese brands own sedans.
- MMR is the strongest predictor of actual selling price — the used car market is efficiently priced around this benchmark.
- Condition, age, and mileage all influence price in expected directions, but no single factor is decisive on its own.
- February is the best month to sell — highest prices and the only month where more transactions close above MMR than below it.
- Neutral colors (black, white, silver, gray) dominate inventory at 69%. Buyers wanting unusual colors will find very thin supply.
- Toyota holds its value best across all age groups. Dodge depreciates fastest among the top 5 brands.
- Geographic concentration in Florida, California, and Georgia reflects the location of major Manheim auction facilities — this is not a random national sample.
- Data quality required significant work: WMI decoding, group-mode imputation, condition scale normalisation, and multiple rounds of odometer cleaning. The final dataset retained 98.7% of raw records.

End of Report

Vehicle Sales Market Analysis • MotoDB Dataset